

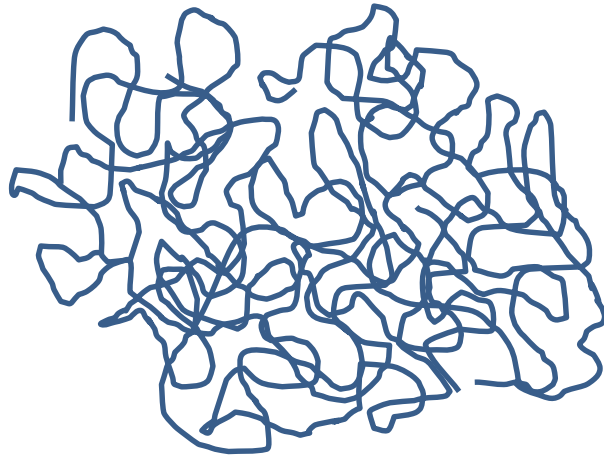


Polymers Crystallization

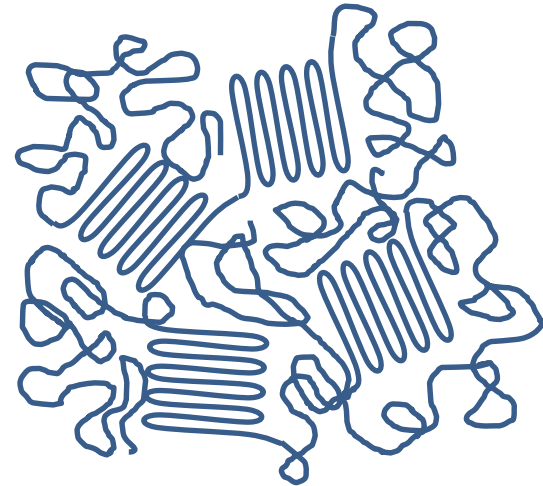
Presented by Dr Adijat 'Wumi Inyang

Polymer crystallization

Polymers are composed of long chain molecules



Polymer molecules are flexible and curl around to make a random coil with no regular structure (**amorphous**) – the molecules become entangled



In some polymers the molecules will form regular arrays in some places with random coils in between – these are **semi-crystalline** polymers

Polymer crystallization

Many thermoplastic polymers and elastomers will crystallize when cooled from the liquid phase (above T_m) to below the melting temperature of the crystalline phase.

- Polymers are very rarely 100% crystalline, usually they are semi-crystalline
- Polymer chains align with each other
- Secondary interactions between polymer molecules
- Will depend on type of side group

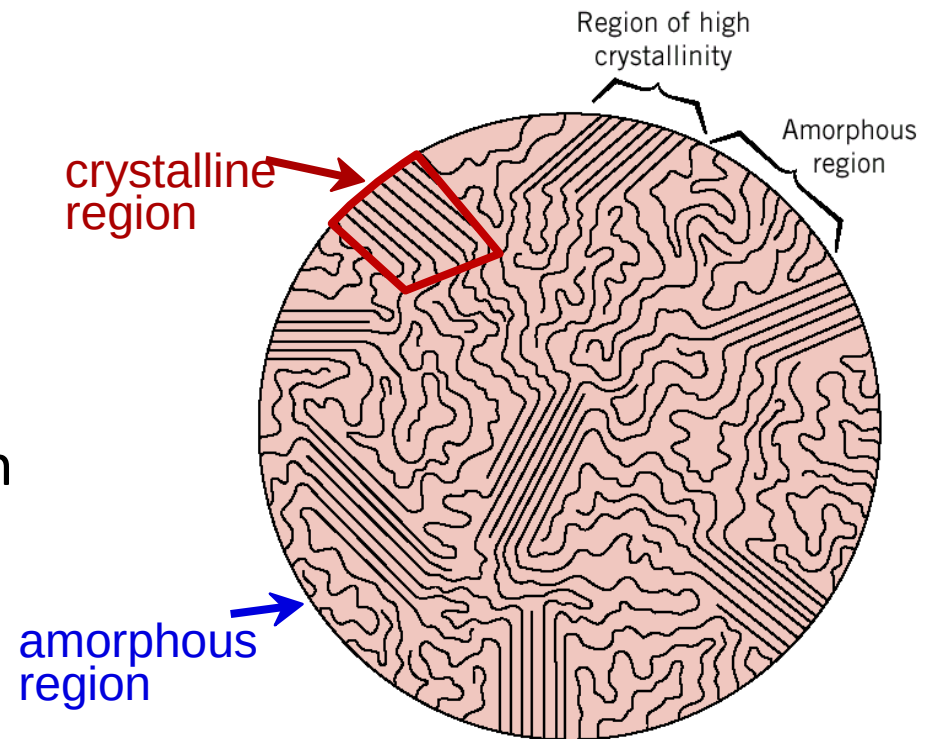
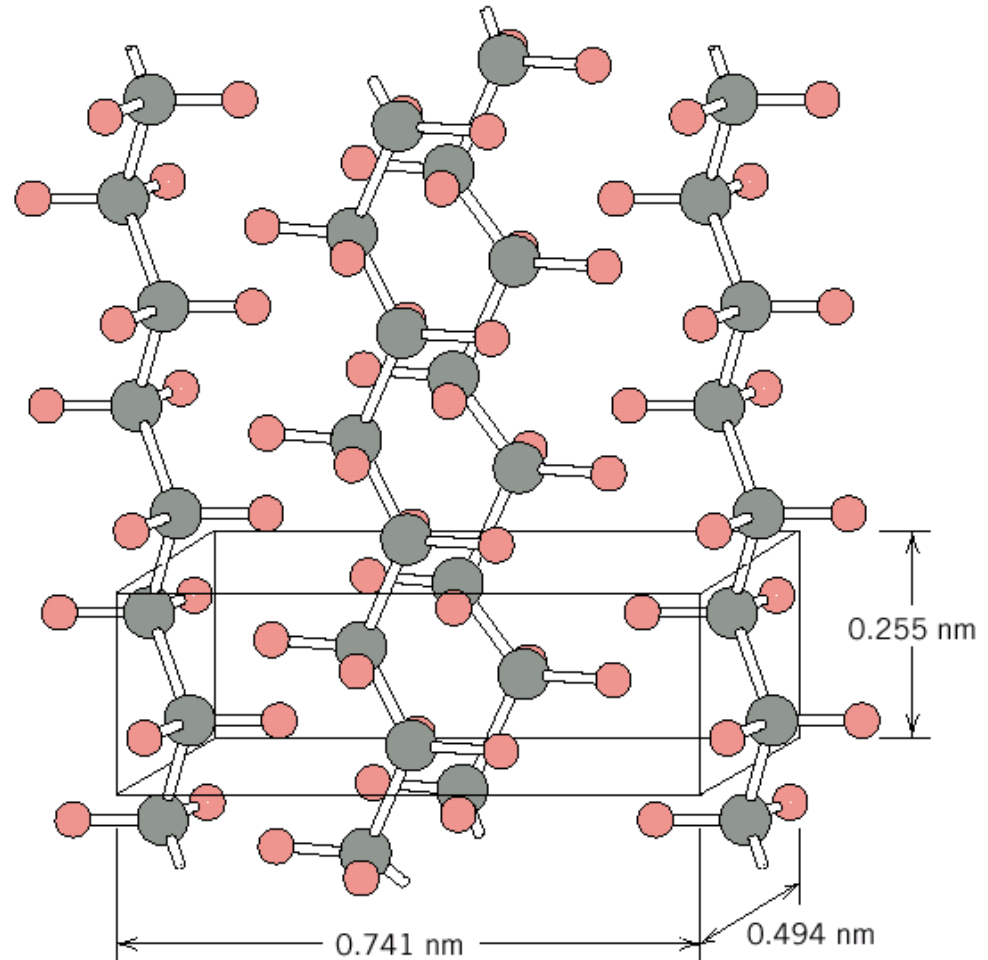


Fig. 14.11, *Callister 6e*. (From H.W. Hayden, W.G. Moffatt, and J. Wulff, *The Structure and Properties of Materials*, Vol. III, *Mechanical Behavior*, John Wiley and Sons, Inc., 1965.)

Polymer crystals

- Ordered atomic arrangements involving molecular chains
- Crystal structures in terms of unit cells
- Example shown
 - **polyethylene unit cell**
- This unit cell is quite simple
- Units cells for polymers can be very complex with many atoms / unit cell



Polymer crystallization

The driving force for crystallization is thermodynamic

The free energy, G , of any system is given by:

$$G = H - TS$$

where H = enthalpy, S = entropy, T = temperature (Kelvin scale)

Nature wants G to be a minimum

There is a reduction in entropy for aligned polymer molecules compared to their usual random coil. This reduction in entropy is countered by the even larger reduction in enthalpy of crystallization (latent heat) , ΔH_m

Therefore, if $\Delta H_m > T_m \Delta S_m$ crystallization will be favoured **to reduce G**

Polymer crystals

- Crystalline regions
 - Regions where the chains align form **crystallites**
 - Thin platelets with chain folds at faces $\sim 10\text{nm}$
 - **Chain folded** structure
 - Imperfect crystals

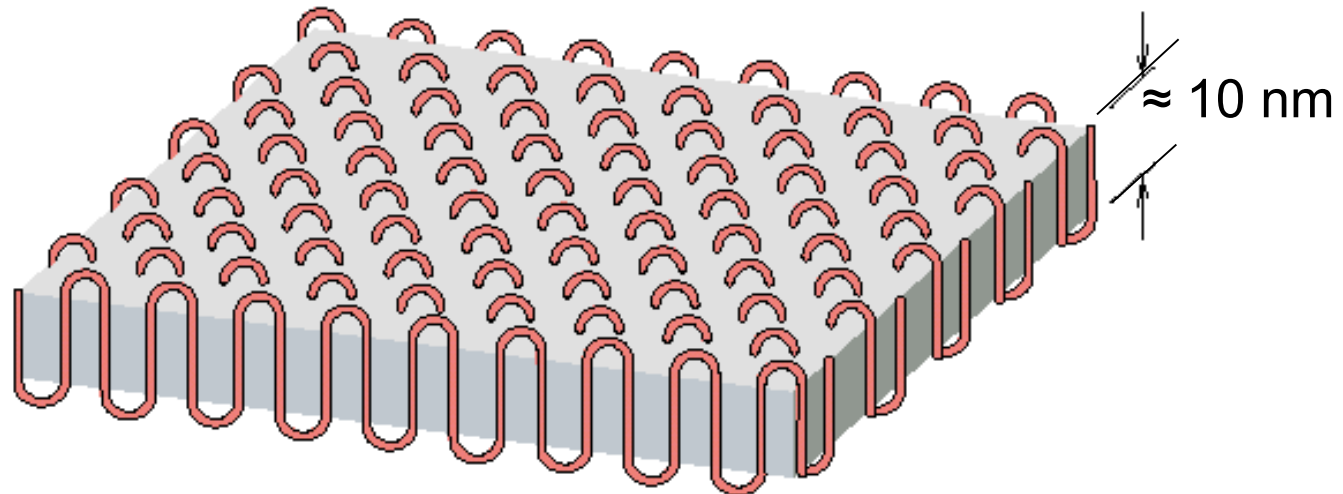
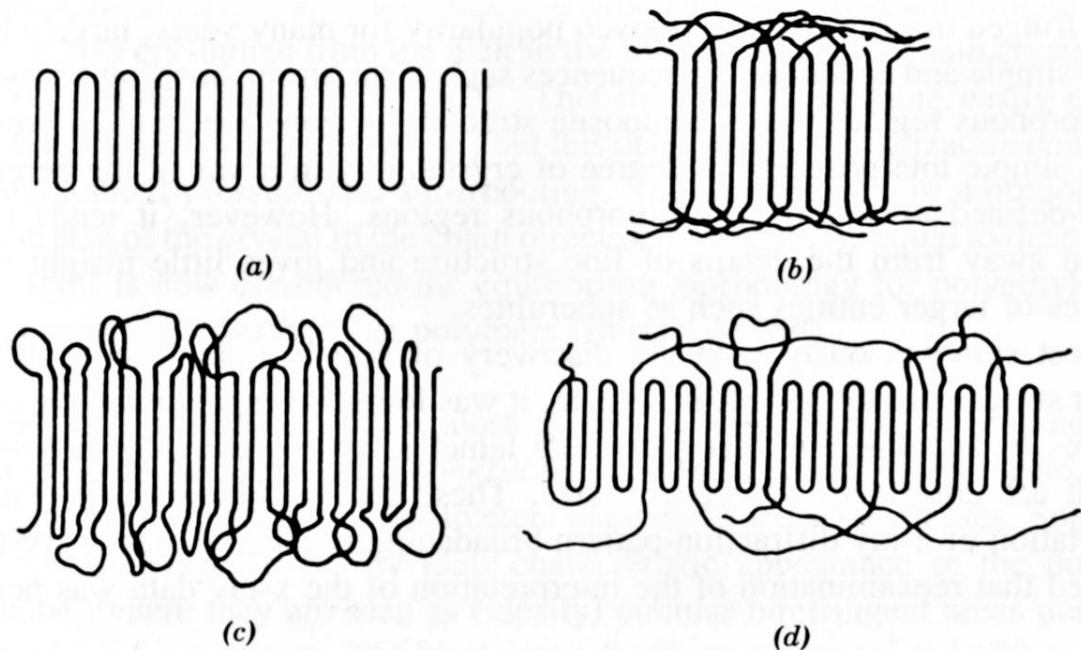


Fig. 5.11, Callister & Rethwisch 9e.

Polymer crystals

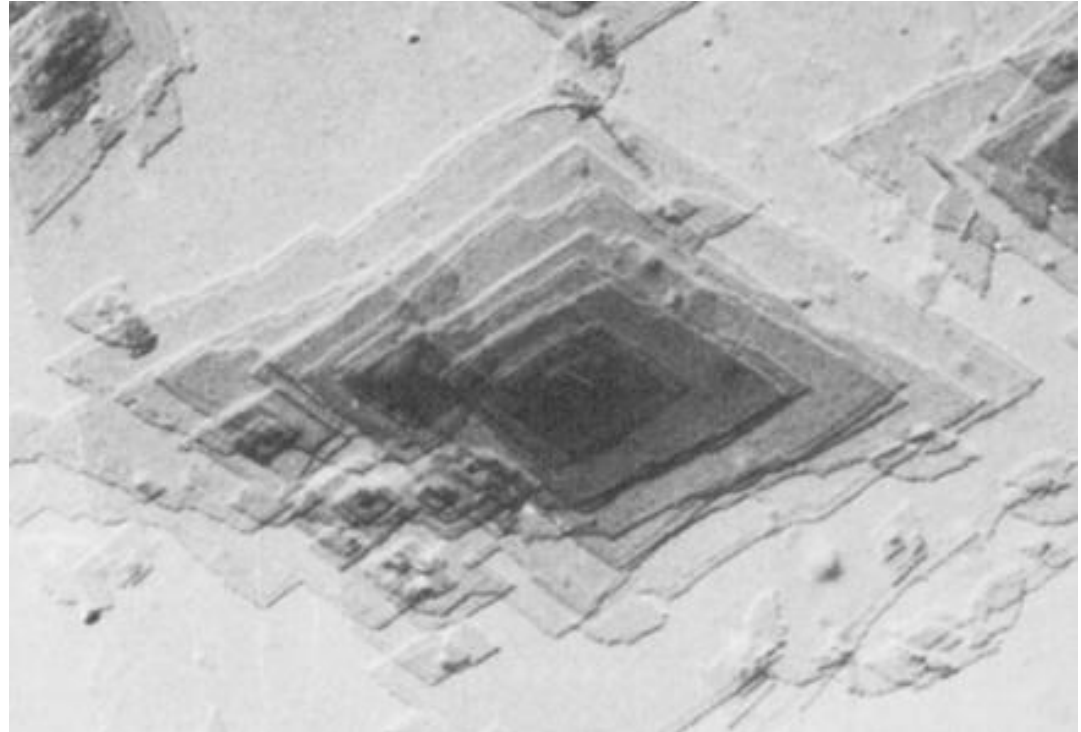
- The polymer chains in the crystallites are **VERY unlikely to be like (a)**



- **They are much more likely to be like (b), (c), or (d)**
- Polymer crystallites are imperfect crystals with many defects
- Annealing above T_g can improve crystal perfection (but not 100%)

Crystallization of polymers

- Scanning Electron Micrograph (SEM) – multilayered single crystals (chain-folded layers) of polyethylene
- Single crystals – only for slow and carefully controlled growth rates



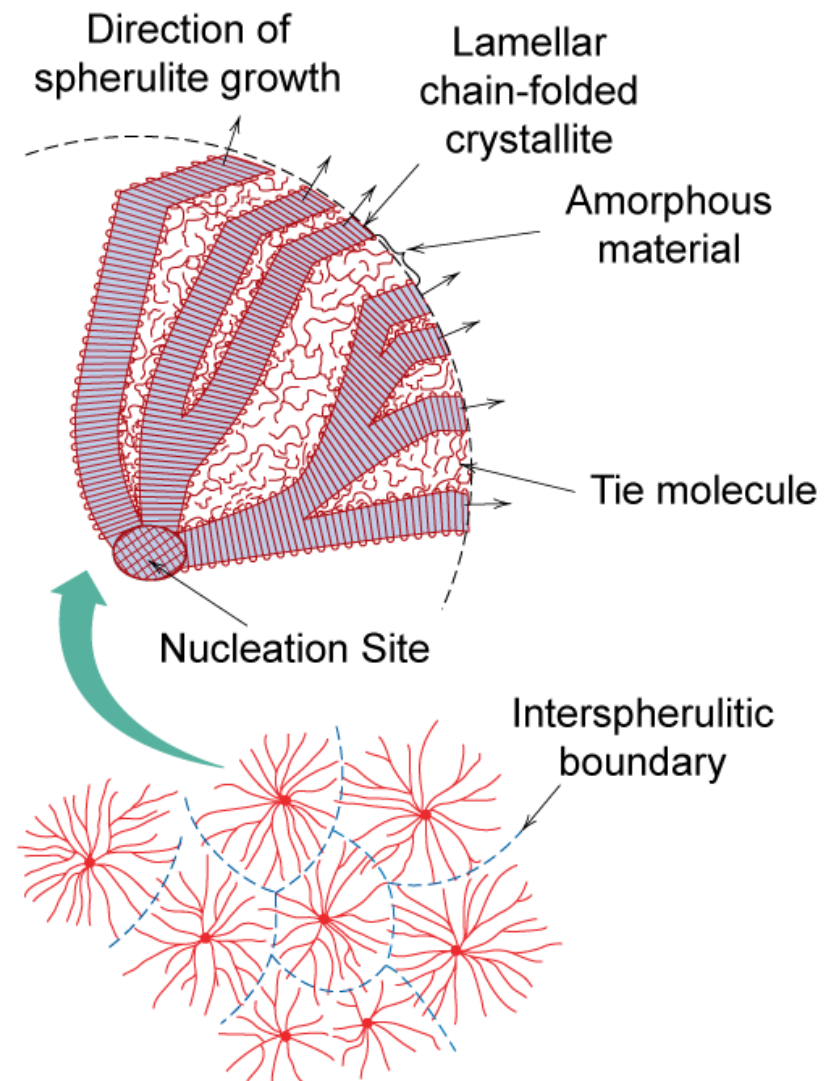
1 μm

Fig. 5.10, *Callister & Rethwisch 9e*.

[From A. Keller, R. H. Doremus, B. W. Roberts, and D. Turnbull (Eds.), *Growth and Perfection of Crystals*. General Electric Company and John Wiley & Sons, Inc., 1958, p. 498. Reprinted with permission of John Wiley & Sons, Inc.]

Crystallization of polymers

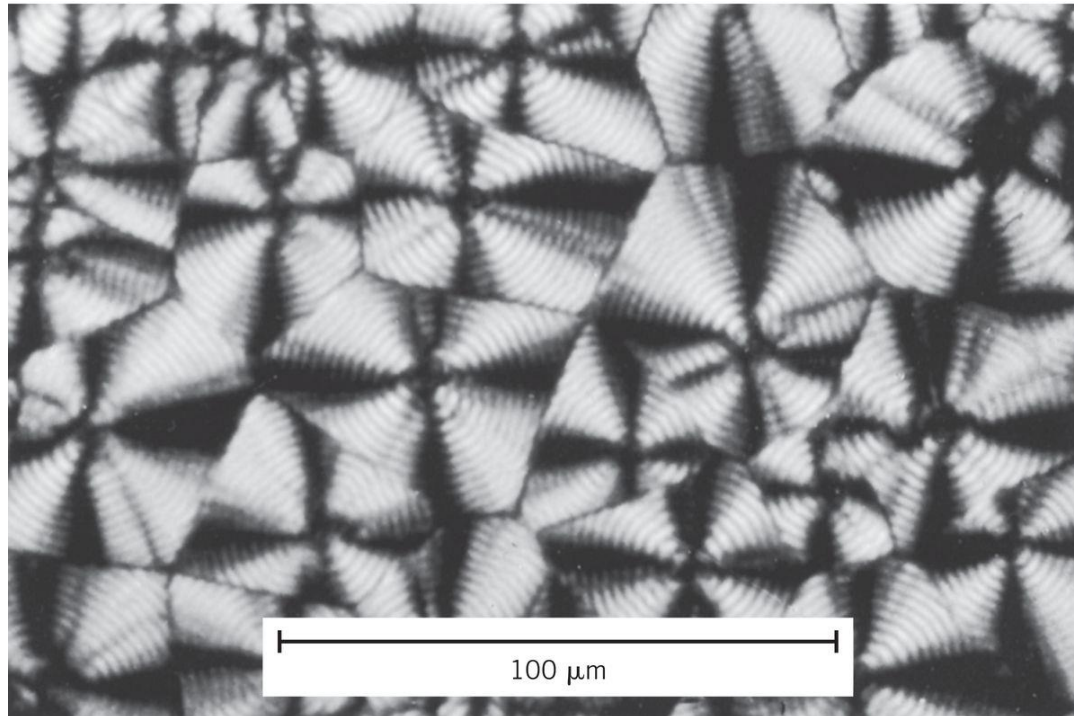
- Some semi-crystalline polymers form **spherulite** structures when crystallizing from the melt
- Alternating chain-folded crystallites and amorphous regions
- Starts from nucleation site and grows outwards until neighbouring spherulites touch
- Spherulite structure for relatively rapid growth rates
- Crystallization encouraged by adding nucleating agents



Crystallization of polymers

The spherulite structure can be seen in **polarized transmitted light microscopy**

- The crystallites cause rotation of the light giving this characteristic pattern



Courtesy F. P. Price, General Electric Company

- See video of polymer crystallization on QMPlus

Crystallization of polymers

Spherulite of polyisoprene,
imaged in the **Transmission Electron Microscope (TEM)**

The electron diffraction
pattern shows that the
sample is crystalline

The polymer was stained
with OsO_4 to give the
contrast between the
amorphous and crystalline
regions (negative stain)

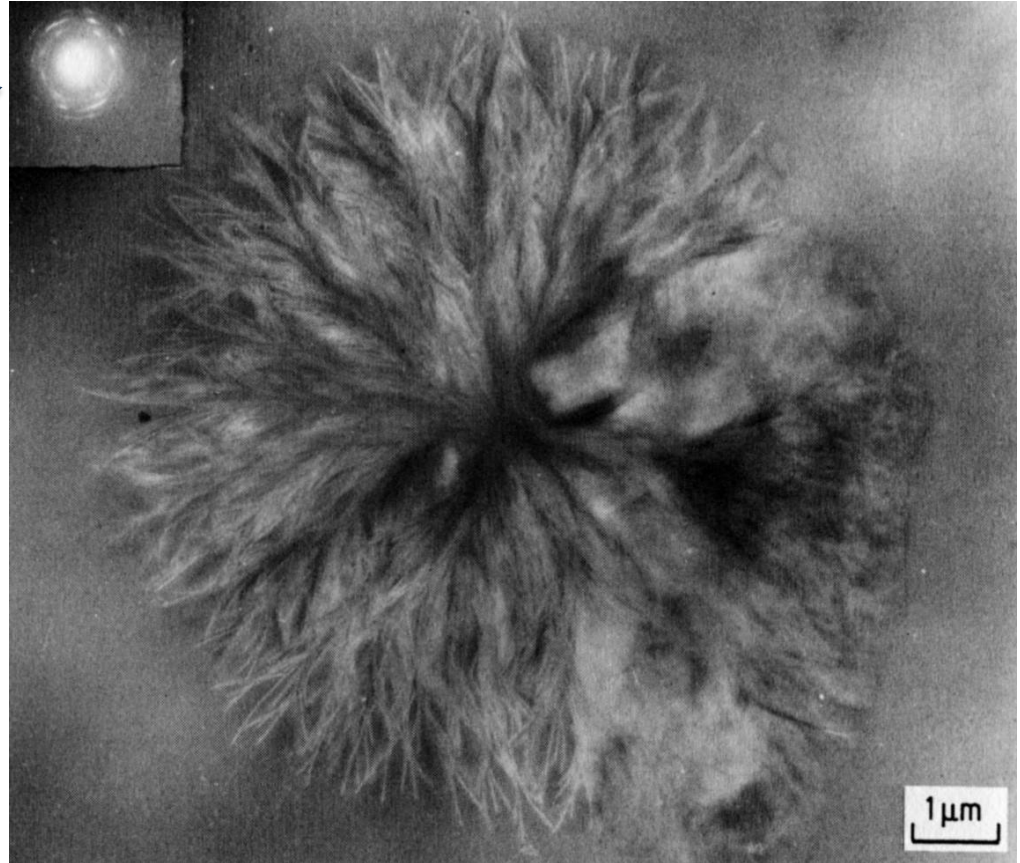


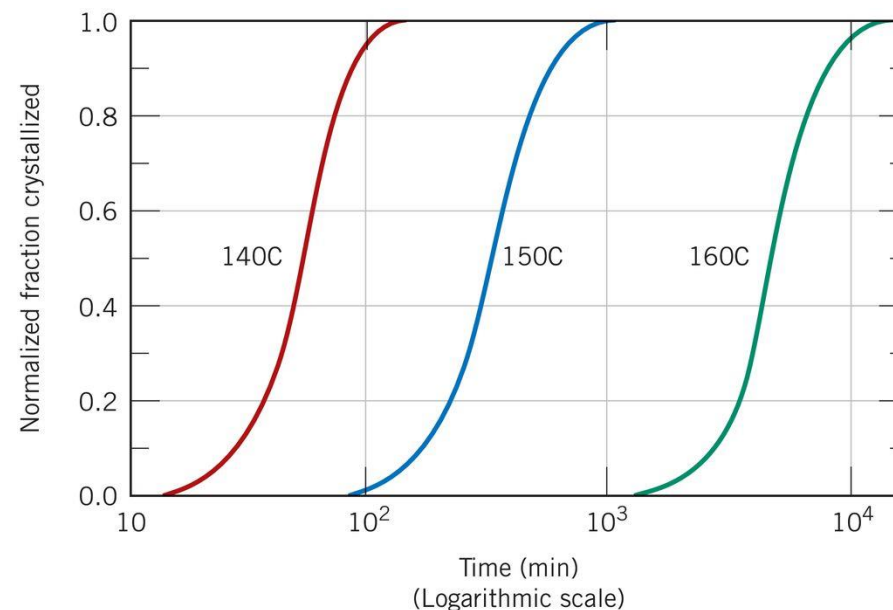
Figure 4.15 R.J. Young Introduction to Polymers
Courtesy of C.K.L.Davies

Rate of crystallization

The rate of crystallization can be described by the Avrami equation

$$y = 1 - \exp(-kt^n)$$

where: k and n are time-independent constants for the crystallizing system



Note: Normalised fraction crystallized = fraction of possible crystallization NOT 100%

Degree of crystallinity

The **degree of crystallinity** in the polymer is an important parameter for the properties of the polymer but not very easy to estimate!

Specific volume method:

$$\psi_c = \frac{V_a - V}{V_a - V_c}$$

where:

ψ_c = volume fraction of crystals in the sample

V_a = volume amorphous (from measurement of fully amorphous material)

V_c = volume pure crystalline (from knowledge of the unit cell)

V = total sample volume (**measured in the experiment**)

Degree of crystallinity

X-ray diffraction:

The intensity of X-ray diffraction peaks are separated (deconvoluted) into those from crystal planes and the amorphous background scattering

The **degree of crystallinity** is then the ratio of the area under the peaks

The crystalline peak width, $B = \Delta 2\theta$, also indicates the **crystallite size**, x , from the Scherrer equation:

$$x = \frac{0.9\lambda}{B \cos \theta}$$

where λ = X-ray wavelength

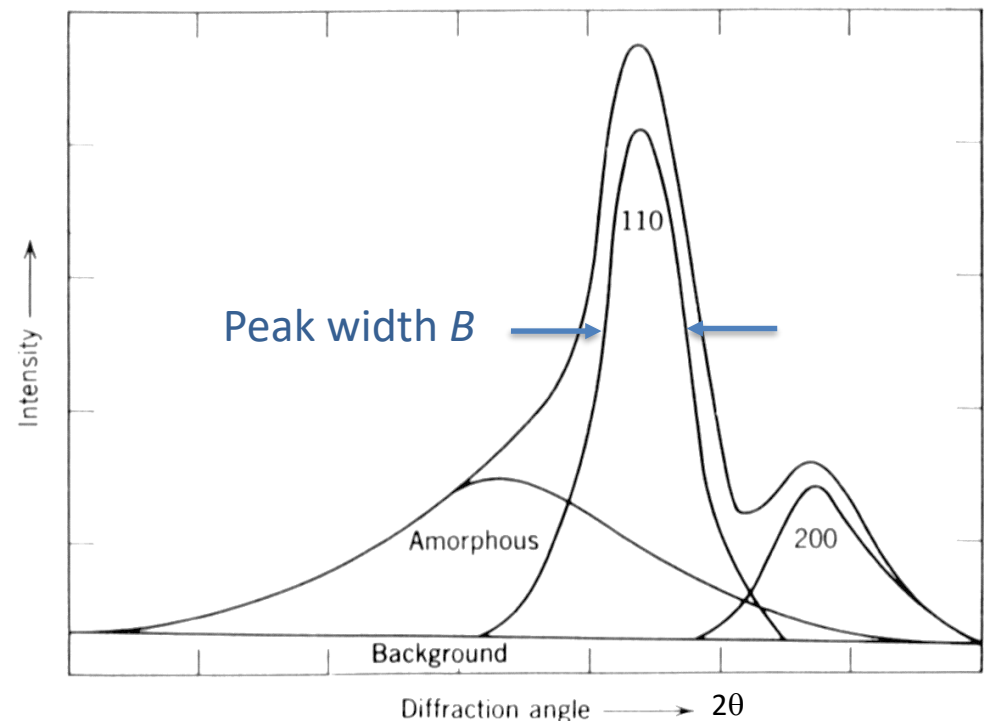


Fig. 10.17, F.W. Billmeyer: *Textbook of Polymer Science* 3e.

Thermal behaviour

Differential scanning calorimetry

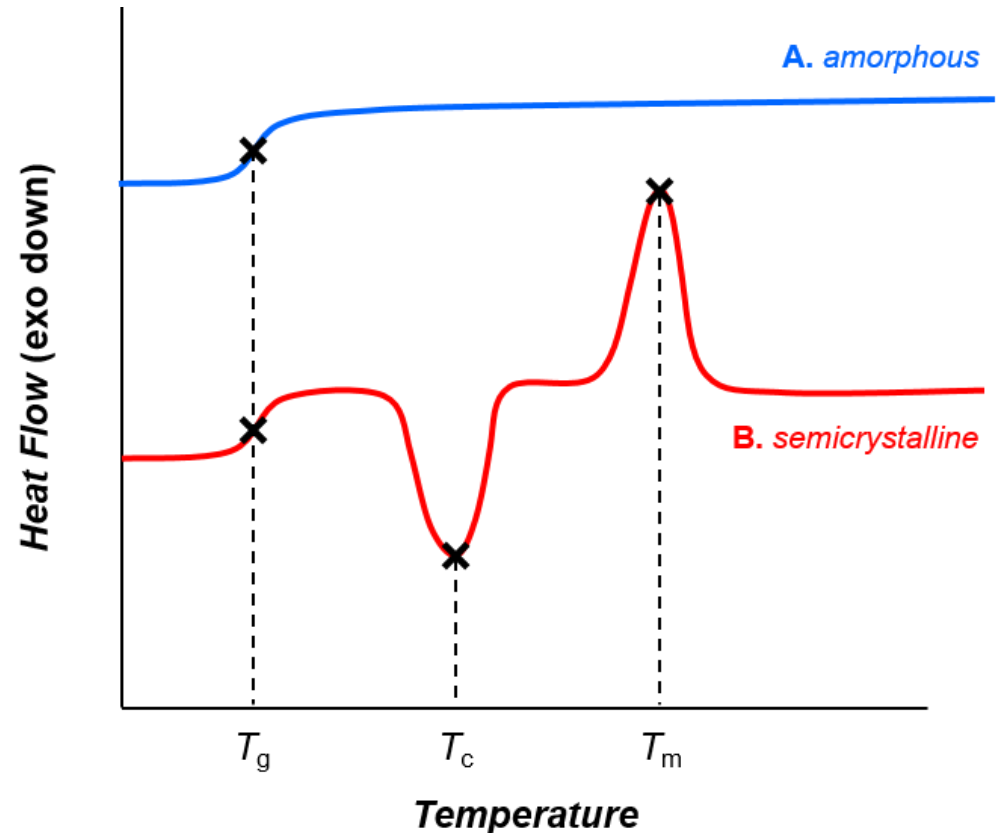
relative heat flow (exothermic down) vs temperature

T_c = crystallization temperature

Crystallization is **exothermic**
(gives out heat)

Crystal melting is **endothermic**
(takes up heat
= latent heat of melting)

Comparison of amorphous and crystalline material provides an estimate of **degree of crystallinity** from the area of the peaks



Mechanical behaviour

Factors that affect the mechanical properties of semi-crystalline polymers

1. Molecular weight

Tensile strength increases with molecular weight for some polymers

2. Degree of crystallinity

Tensile modulus increase with crystallinity

Closely packed, ordered, parallel chains create secondary bonding that makes polymer stronger

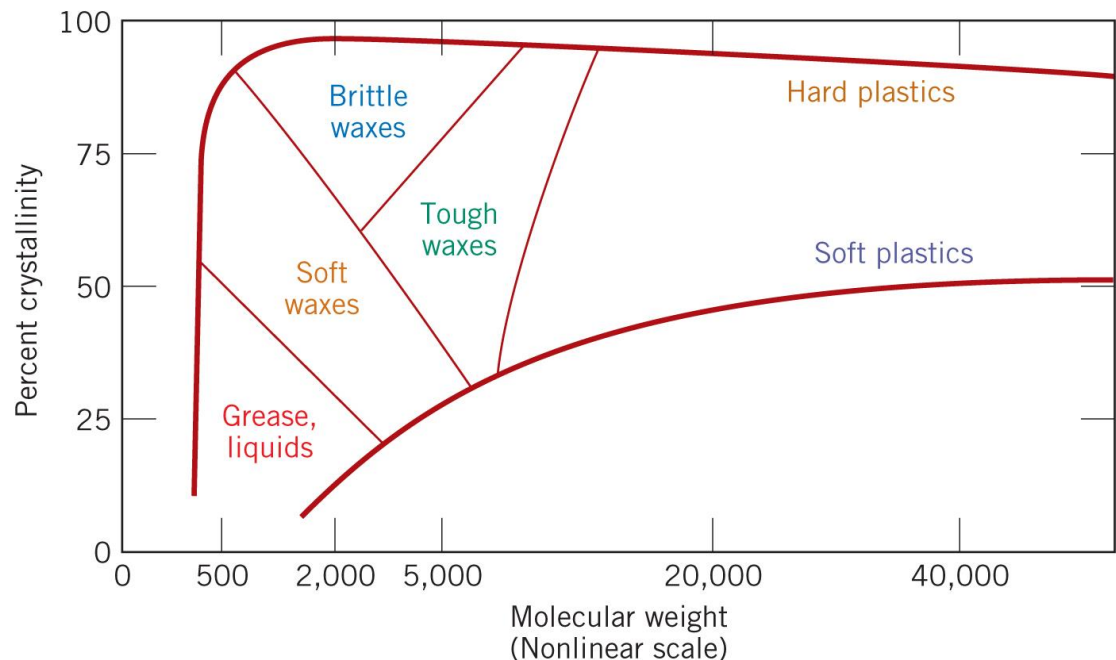


Fig 15.14 Callister & Rethwisch 9e

Summary

- Thermoplastic polymers and elastomers can crystallize by alignment of the polymer chains
- The degree of crystallization is never 100% but some fraction – crystallites with amorphous polymer between
- Crystallization changes the properties of polymers
- Crystallization is affected by polymer processing



Additional online learning material

- Read additional information in Chapter 5, Callister 9e
 - Refresh knowledge from other modules
- Complete the crystallinity in polymers lesson in:
- <https://www.doitpoms.ac.uk/tlplib/polymers/index.php>
- Read papers uploaded in QMPlus
- Complete the Polymer crystallinity quiz in QMPlus
 - Formative assessment